Marvin Chin Yi Kai

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Summary of Qualifications

Full Stack AI Engineer working across the **LLM application layer.** Core stack: **Python, LangChain, LangGraph, FastAPI,** and **vector databases,** with production experience building **RAG pipelines, multi-agent systems,** and **context management** for long-running agents. Experienced connecting LLMs to enterprise systems via **REST APIs, Microsoft Graph API, SAP OData, and MCP servers**. ML foundation in **TensorFlow, PyTorch,** and **on-device model deployment** (PyTorch Mobile, TFLite quantization).

Relevant Experience

Mettler Toledo Selangor, Malaysia
AI Solutions Developer *Apr 2026 - Present*

- Build and maintain **ChatMT Agent**, Mettler Toledo's internal platform for building AI agents: 1,000+ agents built, ~6,000 conversations/day, 1,200 daily users. Own the **agent harness and orchestration layer**: model selection, tools, connectors, system prompts, cost management, and the platform UI/UX.
- Engineer **context management for long-running agents**: recent turns stay verbatim, older history is summarized, oversized tool outputs are auto-summarized, and important facts persist as user memory, preventing context rot and hallucination on long tasks.
- Build the **connectors that let agents act on internal systems**: ServiceNow, Salesforce, SAP, Microsoft 365, and MCP servers, and shipped human-in-the-loop automation agents on top: an **Excel agent** that reconstructs spreadsheets (heuristic + AI detection) as SQL tables over millions of rows, and a **PowerPoint agent** that cut deck assembly from hours to ~40 minutes.
- Added **skills and subagent orchestration**: package a workflow once for any agent to load, and split long-running workflows across multiple agents, each holding only its own context.
- Led the **database and server migration** of MT Translate, Mettler Toledo's text and image translation platform from on-prem to Azure, implementing EasyAuth so any employee can use it without managing accounts.

Deriv.com Cyberjaya, Malaysia
AI Engineer *Mar 2025 - Present*

- Architected and deployed **Agentic RAG chatbots** for finance, compliance, and banking teams using **LangGraph, FastAPI, and vector databases**, supporting advanced features such as **OCR, image captioning, table parsing, contextual chunking, and re-ranking**, reducing manual workload by **40%**.
- Built a **knowledge base management system**, enabling **non-technical admins** to manage documents and AI workflows with **zero-code interaction**.
- Developed a **Multi-Agent Tax DeepResearch System** that automated jurisdictional research and provided **AI-generated market entry strategies**, cutting down research time by **60%** and increasing cross-border decision-making speed.
- Led **UI/UX design** and development for internal platforms like **Finance Hub** and **Compliance Hub**, enabling cross-team adoption of AI tools and consolidating compliance operations under one interface.
- Monitored and optimized production-level AI infrastructure, achieving **99% system uptime**, proactively patching vulnerabilities, and resolving incidents with **<1 hour mean time to recovery (MTTR)**.

Dell Technologies Cyberjaya, Malaysia
Software Engineer Intern *Aug 2023 - Jan 2024*

- Designed and deployed a **time-series forecasting model using Python and LSTM** for edge device compliance, enabling **early detection of security anomalies**.

- Automated **security patch retrieval workflows** using web scraping and scripting tools (Python + PowerShell), reducing manual labor by **~30 hours per month**.
- Contributed to vulnerability remediation, supporting compliance efforts that achieved **99% endpoint compliance rate** across monitored devices.

Education

Universiti Malaya, *Faculty of Computer Science & Information Technology* Kuala Lumpur, Malaysia
Bachelor of Computer Science (Artificial Intelligence). CGPA: 3.92/4.00 *Nov 2025*

- **First Class Honours, Penang Future Foundation Recipient**
- Relevant Coursework: Deep Learning, Computing Mathematics, Computer Vision, Natural Language Processing

Technical Skills & Projects

Programming: Python, JavaScript, Kotlin, Java, PowerShell
AI/ML Frameworks: TensorFlow, PyTorch, Scikit-learn, OpenCV, LangChain
Cloud & Tools: AWS, Azure, Terraform, Langfuse, FastAPI
Design & Prototyping: Shadcn UI, Figma, Photoshop, Illustrator, Canva

Mobile Application for Malaysian Sign Language Translation *Mar 2024 - Feb 2025*

- Partnered with MFD to build LSTM-powered mobile app achieving **93% accuracy across 100+ gestures**, improving accessibility for the deaf community.

AI-Powered Feedback Analysis Website for Events *Mar 2024 - Feb 2025*

- Built sentiment analysis platform (LLMs + NLP) for SEKRUM, enabling **faster, data-driven event decisions**.

Leadership

Universiti Malaya Computer Society (PEKOM) Kuala Lumpur, Malaysia
Vice President *Mar 2023 - Jun 2024*

- Expanded club activities by **700% per session**; forged partnerships with societies, companies, and government agencies.

Universiti Malaya Tenth Residential College Kuala Lumpur, Malaysia
Head of Multimedia, Student Facilitator *Jun 2022 - Jun 2024*

- Led orientation for **600+ new students** and directed events recognized with **Best Innovative Project Award (2024)**.